

Textbook EIA Work Progress Logging

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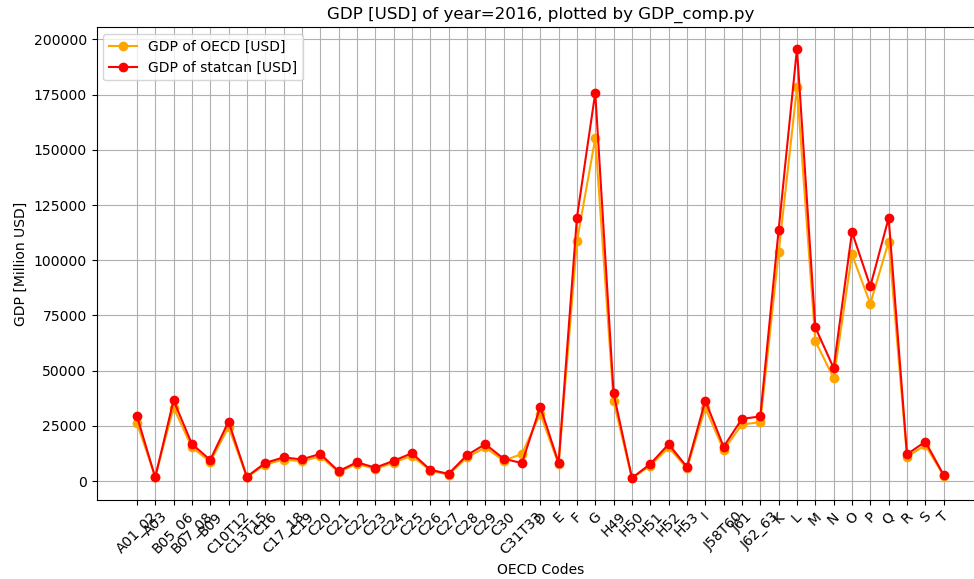
1 Uploading data from OECD and Statcan and Integrating Them

1.1 File *GDP_comp.py*

In *GDP_comp.py* I upload OECD data and Statcan data and integrate them by

1. Converting Statcan from CAD to USD
2. Mapping the Statcan sectors (more detailed) to OECD sectors

GDP_comp.py is the same as *func_upload_data.py* where I upload OECD and statcan data into *income_multipliers*.



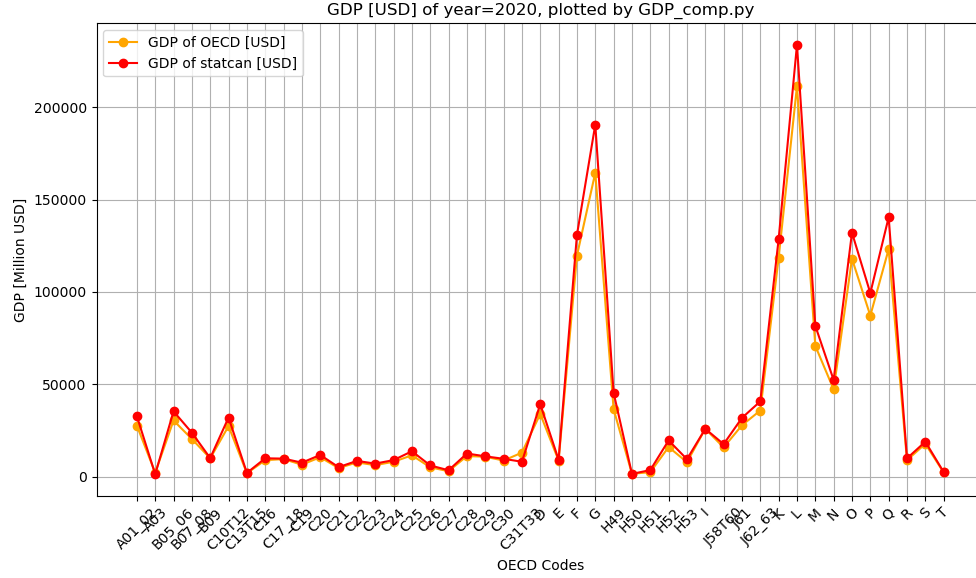


Figure 2: Value Added by Sector (StatCan, 2020)

1.2 Plotting T as function of year

plot_T_vs_time.py plots T of the 6 ICT sectors as function of year. Sector G (wholesaling) has variability in T .

1.3 Comparing OECD and statcan: GDP, II and output

Using *OECD_statcan_comp.py* comparing GDP and II and output between OECD and statcan.

1.4 Uploading all data from statcan - not just GDP

In *read_statcan.py* I upload all data from statcan - not just GDP. I need the “compensation of employees” for the closed model. I updated *func_data_upload.py* to *func_data_upload2.py* to include all data of statcan.

1.5 Output prediction

In *output_multipliers_all_sectors.py* I plot the predicted output versus the real output. The simple model gives very exact results.

Simple model: the predictions are accurate and consistent (see Fig. 5, 6) even in 2020 which shows the biggest divergence of T (graph not shown).

The prediction of the closed model is not very accurate. Possible reasons include:

1. A mistake in the implementation of the model;
2. Inaccuracies in the conversion from CAD to USD (PPP uploaded versus PPP experimental);

Column Values in T per Sector, by Year

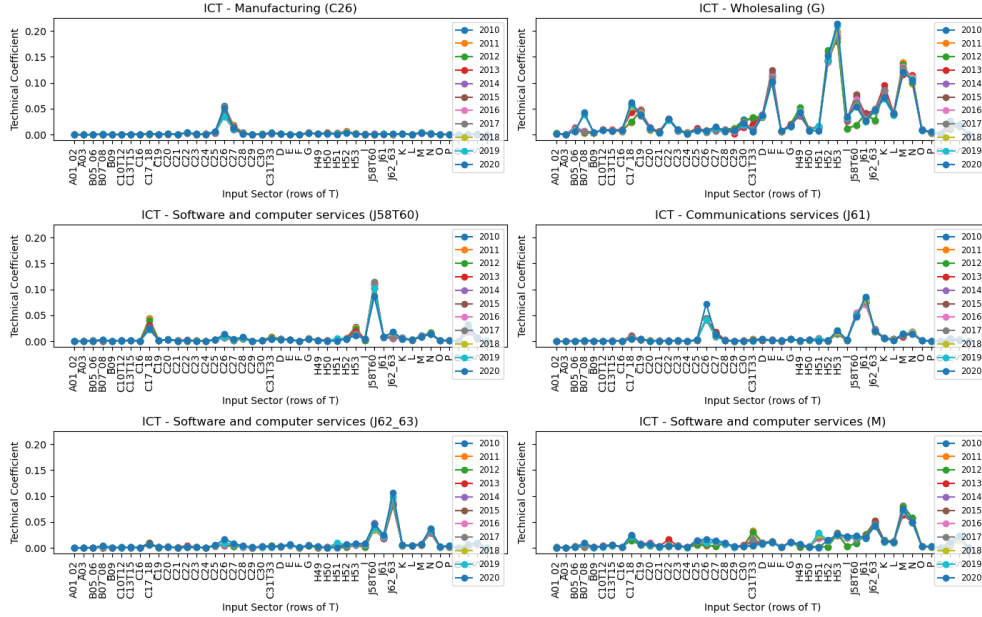


Figure 3: Value Added by Sector (StatCan, 2010-2020)

3. Errors introduced by the unification of OECD and StatCan data that are caused by the different data collection methods.

1.6 Summary of files

upload_data.py - this is uploading only GDP from statcan, and not a function. The early version of *func_upload_data2.py* *output_multipliers_all_sectors.py* where I calculate multipliers and plot predictions *OECD_statcan_comp.py* comparing GDP and II and output between OECD and statcan.

1.7 Correction of final demand vector

On 27.4.25 I realized the final demand vector I use for prediction - $f_{cdf_{year2}}$ in the code - should include the column HFCE from OECD. That is because the last entry of the output vector x is 0 and it shouldn't be -it should be either 1 or -1 , depending on which side of the equation it is (with f or on the other side). Briefly,

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

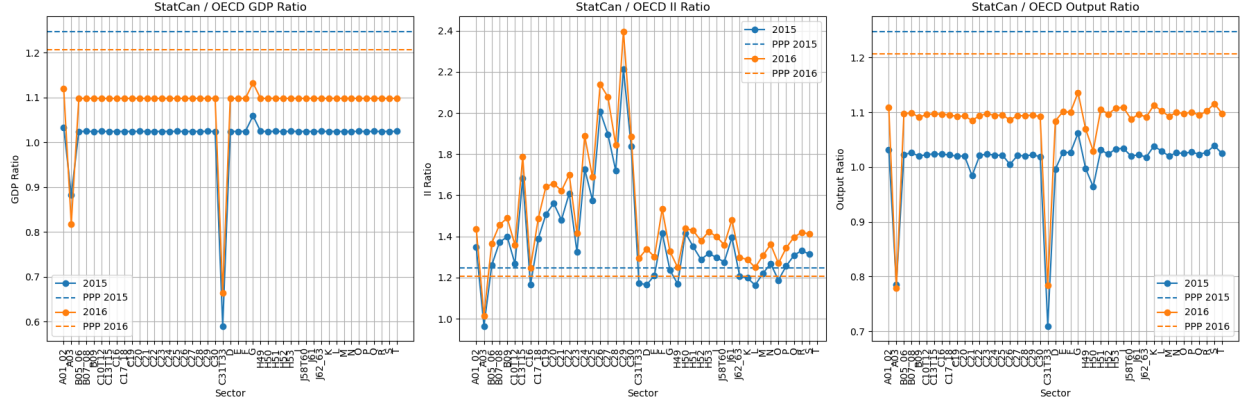


Figure 4: Plots of the ratio Statcan/OECD (CAD/USD) of Output, Intermediate Input and GDP. Also plotted are PPP, the CAD to USD conversion ratios uploaded form Statcan. It is clear that the actual ratios of output and GDP, which should reflect the PPP are far from these theoretic values. Also clear that the Output and GDP are generally similar, except for the sectors A03, C31T33, G, H50.

Now, if we split f to $f^1 + f^2$ and we insert f^1 into A we get,

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & f_1^1 \\ a_{21} & a_{22} & \cdots & a_{2n} & f_1^1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & f_n^1 \\ e_1 & e_2 & \cdots & e_n & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ 0 \end{bmatrix} + \begin{bmatrix} f_1^2 \\ f_2^2 \\ \vdots \\ f_n^2 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ 0 \end{bmatrix}$$

The zeros at the bottom of f and x are given by OECD.

This means that when multiplying A by x , f^1 is actually multiplied by 0 and does not enter the calculation. If it were 1 instead of 0 at the bottom of x , f^1 would have entered the calculation. Since x is dictated by OECD and can't be changed, I have got to add f^1 to f^2 to have the correct calculation:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & f_1^1 \\ a_{21} & a_{22} & \cdots & a_{2n} & f_1^1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & f_n^1 \\ e_1 & e_2 & \cdots & e_n & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ 0 \end{bmatrix} + \begin{bmatrix} f_1^1 \\ f_2^1 \\ \vdots \\ f_n^1 \\ 0 \end{bmatrix} + \begin{bmatrix} f_1^2 \\ f_2^2 \\ \vdots \\ f_n^2 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \\ 0 \end{bmatrix}$$

Adding HFCE to the final demand vector, I get a nice agreement between prediction and reality, see fig. 9. (plotted by *C26_multipliers.py*)

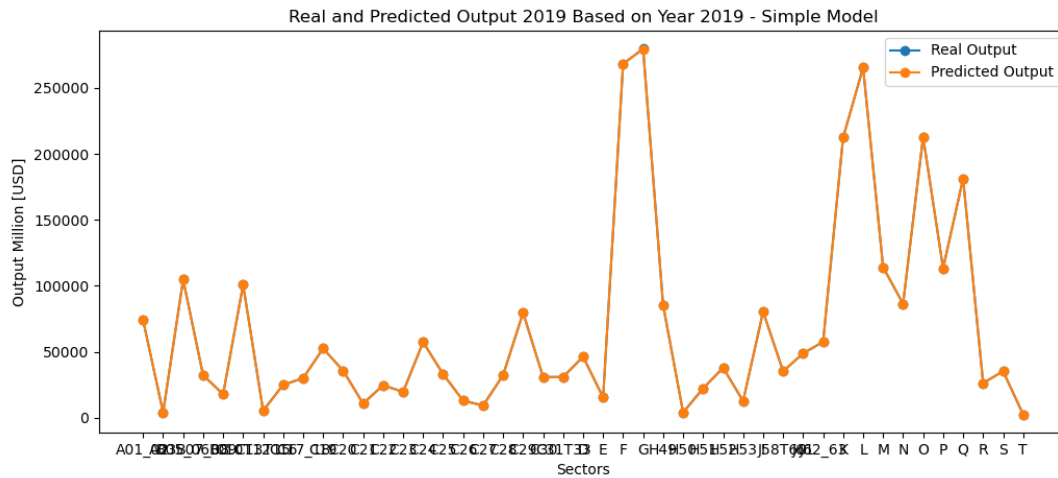


Figure 5: Validation of the model, 2019 predicted output and 2019 real output

Statcan Code	Description	OECD code
A	Agriculture, forestry and fishing	-
A01	Crop and animal production, hunting and related service activities	A01_02
A02	Forestry and logging	A01_02
A03	Fishing and aquaculture	A03
B	Mining and quarrying	-
B05	Mining of coal and lignite	B05_06
B06	Extraction of crude petroleum and natural gas	B05_06
B07	Mining of metal ores	B07_08
B08	Other mining and quarrying	B07_08
B09	Mining support service activities	B09
C	Manufacturing	-
C10	Manufacture of food products	C10T12
C11	Manufacture of beverages	C10T12
C12	Manufacture of tobacco products	C10T12
C10T12	Manufacture of food products; beverages and tobacco products	C10T12
C13	Manufacture of textiles	C13_15
C14	Manufacture of wearing apparel	C13_15
C15	Manufacture of leather and related products	C13_15
C16	Manufacture of wood and of products of wood and cork, except furniture	C16_18
C17	Manufacture of paper and paper products	C16_18
C18	Printing and reproduction of recorded media	C16_18
C19	Manufacture of coke and refined petroleum products	C19
C20	Manufacture of chemicals and chemical products	C20_21
C21	Manufacture of basic pharmaceutical products and pharmaceutical preparations	C20_21
C22	Manufacture of rubber and plastics products	C22_23
C23	Manufacture of other non-metallic mineral products	C22_23
C24	Manufacture of basic metals	C24_25
C25	Manufacture of fabricated metal products, except machinery and equipment	C24_25
C26	Manufacture of computer, electronic and optical products	C26_27
C27	Manufacture of electrical equipment	C26_27

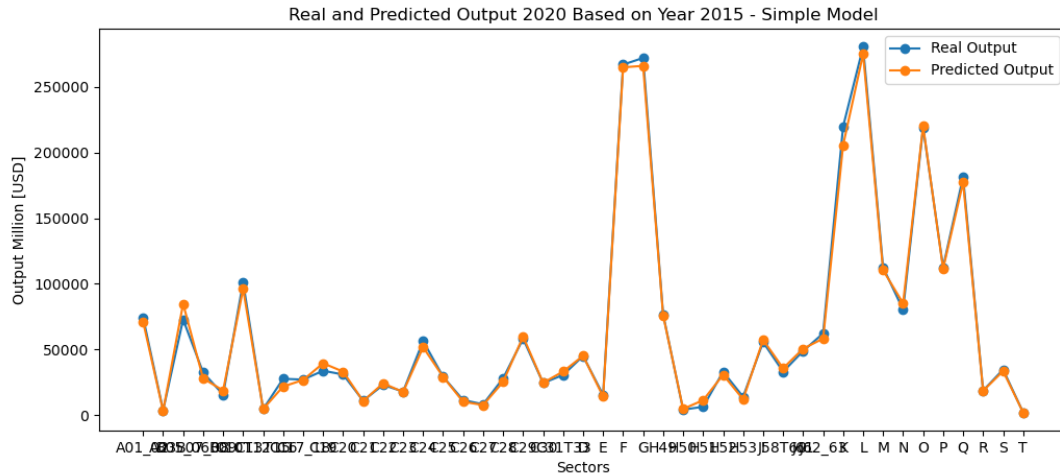


Figure 6: Predicted output of 2020 based on 2015, and real output 2020 plotted by *output_multipliers_all_sectors.py*

Statcan Code	Description	OECD code
I	Accommodation and food service activities	I
I55	Accommodation	
I56	Food and beverage service activities	
J	Information and communication	?
J58	Publishing activities	
J59	Motion picture, video and television programme production, sound recording and music publishing activities	
J60	Programming and broadcasting activities	J58T60
J61	Telecommunications	J61
J62	Computer programming, consultancy and related activities	
J62.63	Computer programming, consultancy, and information service activities	J62.63
J63	Information service activities	
K	Financial and insurance activities	K
K64	Financial service activities, except insurance and pension funding	
K65	Insurance, reinsurance and pension funding, except compulsory social security	
K66	Activities auxiliary to financial service and insurance activities	
L	Real estate activities	L
L68B	Real estate activities excluding imputed rents	
M	Professional, scientific and technical activities	M
M69	Legal and accounting activities	
M69_70	Legal and accounting activities; activities of head offices; management consultancy activities	
M70	Activities of head offices; management consultancy activities	
M71	Architectural and engineering activities; technical testing and analysis	
M72	Scientific research and development	
M73	Advertising and market research	
M74	Other professional, scientific and technical activities	
M74_75	Other professional, scientific and technical activities; veterinary activities	
N	Administrative and support activities	N

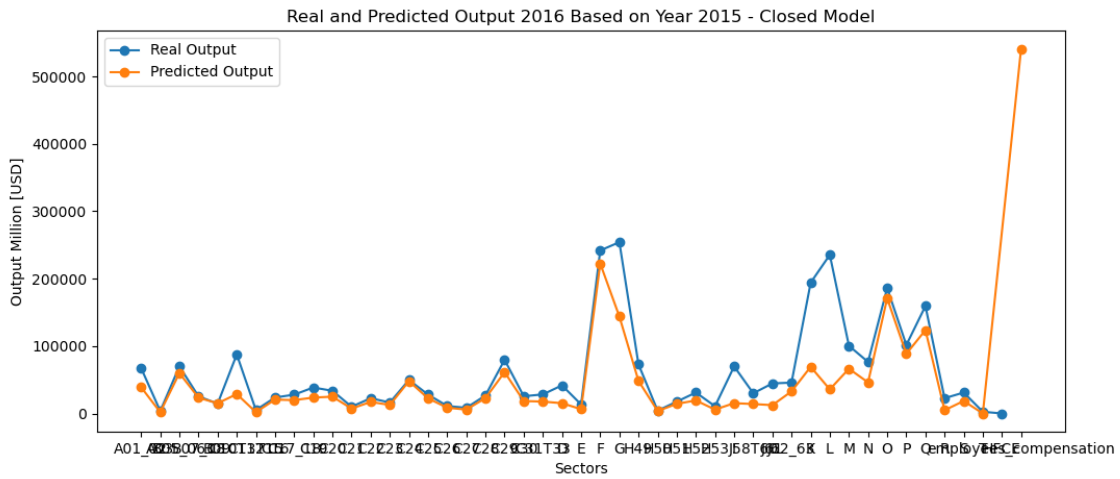


Figure 7: Closed model with PPP uploaded from statcan. Plotted by *output_multipliers_all_sectors.py*

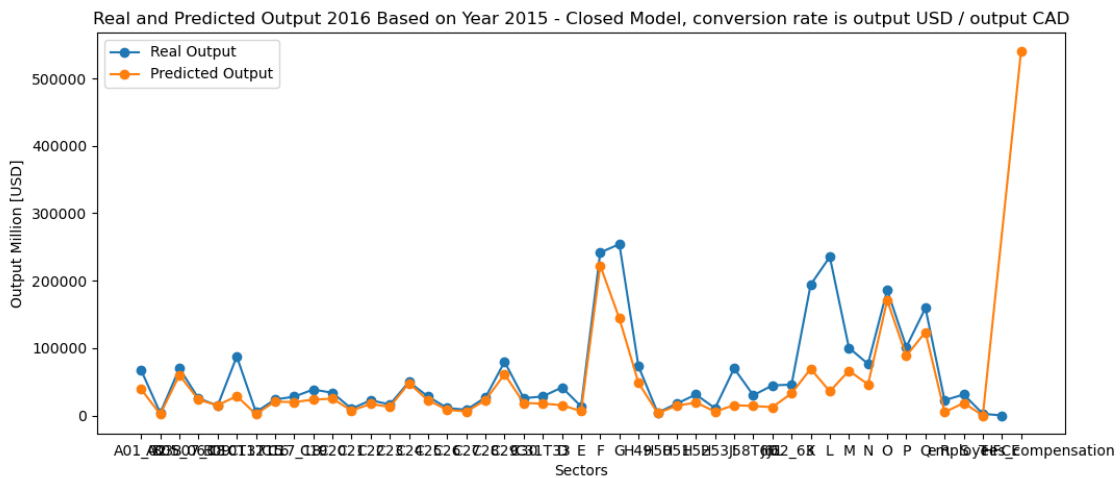


Figure 8: Closed model with experimental PPP - a rough estimation taken from the comparison of OECD GDP and statcan GDP. Plotted by *output_multipliers_all_sectors.py*

Closed Model

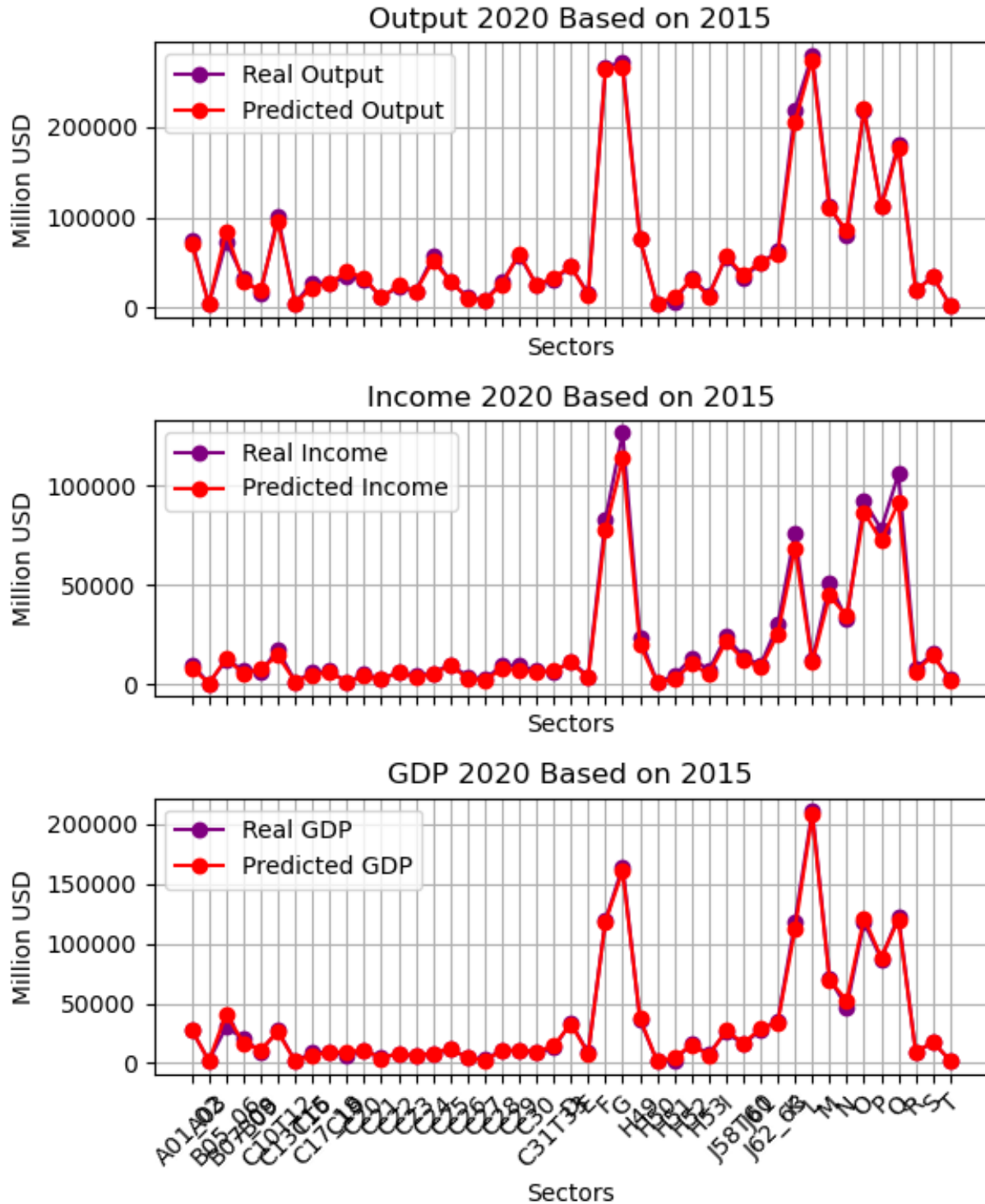


Figure 9: After I corrected final demand to include the HFCE vector the results of the closed model are excellent. Plotted by *C26_multipliers.py*